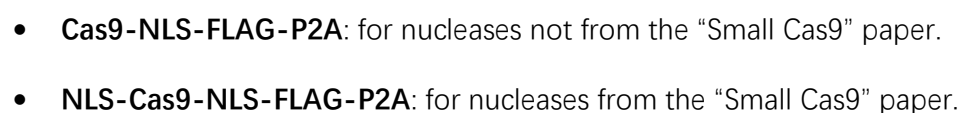


Target context sequence (raw): the original target context sequence taken directly from the non-target DNA strand in the source data. It includes

- Spacer sequence:** the padded version of the raw spacer sequence, with 'N' padding added to the 5' and 3' ends. The padding extends the total length of the spacer sequence to 42 nt, ensuring alignment with the target context sequence.

- **5' context + target:** 27 nt
- **PAM + 3' context:** 15 nt



In these names, **NLS** refers to a nuclear localization signal, **FLAG** refers to the FLAG protein tag, and **P2A** is a type of 2A peptide.

Nuclease: the Cas9 nuclease contained within each variant.

gRNA scaffold: the scaffold of the sgRNA, consisting of the repeat-anti-repeat loop and other stem loops.

Day: the timepoint at which indel frequencies were measured. After introducing Cas9 into cells via transfection (for “Wild SpCas9,” “xCas9_NG,” and “DeepHF” studies) or transduction (for “SpCas9,” “Small Cas9,” “Base Editor,” and “Sniper” studies), cells were harvested and indel frequencies were analyzed by deep sequencing.

tRNA feature: indicates whether tRNA-associated processing happened. If so, the tRNA-N₂₀ sgRNA was cleaved to yield an N₂₀ sgRNA (for more details, refer to Supplementary Figure 3 of the “SpCas9” study).

Study: the source study from which the data was obtained. Each study has an abbreviated name

- **Wild SpCas9** - [SpCas9 activity prediction by DeepSpCas9, a deep-learning model with high generalization performance](#)
- **DeepHF** - [Optimized CRISPR guide RNA design for two high-fidelity Cas9 variants by deep learning](#)
- **xCas9_NG** - [High-throughput analysis of the activities of xCas9, SpCas9-NG and SpCas9 at matched and mismatched target sequences in human cells](#)
- **SpCas9** - [Prediction of the sequence-specific cleavage activity of Cas9 variants](#)
- **Small Cas9** - [Massively parallel evaluation and computational prediction of the activities and specificities of 17 small Cas9s](#)
- **Base Editor** - [Deep learning models to predict the editing efficiencies and outcomes of diverse base editors](#)
- **Sniper** - [Sniper2L is a high-fidelity Cas9 variant with high activity](#)

Library: the lentiviral library used for the data point.

Table: the supplementary table number from the publication containing the data.

Sheet: the sheet in the supplementary table where the data can be found.

src_idx: row number in the sheet corresponding to the specific data point.

n_data: number of times the experiment was repeated for the same guide-target-scaffold-Cas9 combination.

Partition: indicates whether the data point was used in deep learning, and if so, whether it was part of the training or test set, or which fold it belongs to.

Barcode: a unique identifier for each experiment, allowing identification of individual experiments, especially where some guide-target pairs were included multiple times for reliable data recovery.

Background subtracted indel frequencies (%): This is represented as a list of lists of floats, where each inner list corresponds to data from a single study. Each float in these lists is capped at 100 but may occasionally fall slightly below zero due to experimental error.

Mean background subtracted indel frequency (source, %): a list of floats, where each float represents the average value of the **Background-subtracted indel frequencies (%)** for a corresponding list.

Mean background subtracted indel frequency (%): This is a single float obtained through a weighted average. The weights are the integers in **n_data**, and the values are the floats in **Mean background-subtracted indel frequency (source, %)**. It is computed as the dot product of these two vectors divided by the sum of the weights.